



DATA MODELLING & MANAGEMENT

DAL8 SQ A3.3 – WORKSHEET

APPLY STAGE



Goal: Design, build and manage a robust data solution from source to secure use.

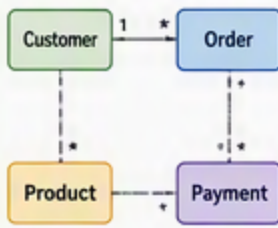
Plan Design Build Manage Evaluate

1 CONDUCT RESEARCH ON DATA MODELLING & MANAGEMENT WITHIN YOUR ORGANISATION

K&Ps: K3, P1, P3 (Part 1); K1, K7, K8 (Part 2)

- Data models
- Tools & technologies
- Standards & best practices
- Data structures
- Naming conventions
- Documentation & resources

EXAMPLE DATA MODEL



TOOLS & TECHNOLOGIES

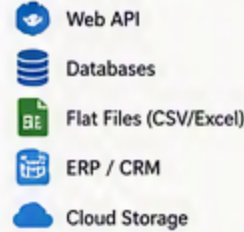


2 CONDUCT RESEARCH ON DATA SOURCES

K&Ps: K1, K6 (Part 1), K9, K10

- Identify data sources
- Types of information
- Due diligence
- Data gathering methods
- Batch processing
- Streaming
- Data import mechanisms
- Screenshots of ETL using PowerQuery

DATA SOURCES



ETL WITH POWERQUERY



3 IDENTIFY REFERENCE DATA TABLES

K&Ps: K11 (Part 1)

- Identify reference data tables
- List of possible reference tables and attributes
- Reference Tables:

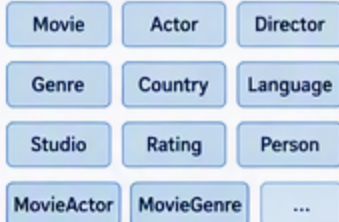
TABLE	KEY ATTRIBUTES
Dim_Genre	GenreID (PK), GenreName, Description
Dim_Rating	RatingID (PK), Rating, Description
Dim_Country	CountryID (PK), CountryName
Dim_Language	LanguageID (PK), LanguageName
Dim_Date	DateKey (PK), FullDate, Day, Month, Year

4 IDENTIFY DATA ENTITIES

K&Ps: K7, K8 (Part 1); K2 (Part 2)

- Find data entities
- List of entities for Movies database
- Entities:
- JSON vs Excel
- Your preferences

MOVIES DATABASE – ENTITIES



JSON (FLEXIBLE)



Good for nested data, APIs, semi-structured data.

EXCEL (STRUCTURED)



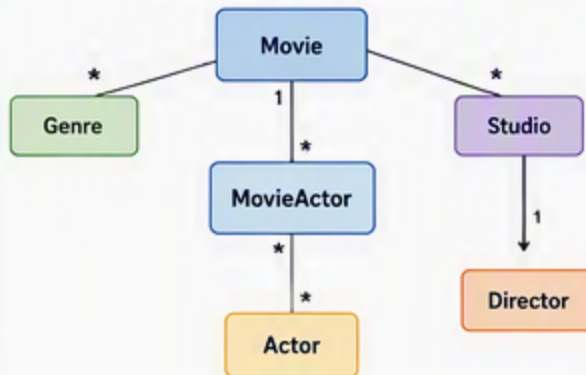
Best for tabular data, easy analysis & reporting.

PREFERENCE: Use JSON for data exchange; Excel for reporting & analysis.

5 CREATE AN ENTITY-RELATIONSHIP (ER) DIAGRAM

K&Ps: K4, K5, P4 (Part 1); K5, K11 (Part 2)

ER DIAGRAM (MOVIES DATABASE)



1 = One * = Many

6 CREATE A BASIC DESIGN DOCUMENT

K&Ps: K2, P5, P7, P8 (Part 1);
- K4, K6, K9, K10, P1, P3 (Part 2)

DATA DICTIONARY (MOVIES DATABASE)

TABLE	COLUMN	TYPE	KEY	DESCRIPTION
Movie	MovieID	INT	PK	Unique movie identifier
	Title	VARCHAR(200)		Movie title
	ReleaseDate	DATE		Release date
	GenreID	INT	FK	References Genre
...	...	...	...	...



Purpose



Scope



Assumptions



Business Rules

7 COMPOSE CREATE STATEMENTS

K&Ps: P9, P10 (Part 1); P2 (Part 2)

SQL CREATE STATEMENTS (DDL)

```
CREATE TABLE Movie (  
  MovieID INT IDENTITY(1,1) PRIMARY KEY,  
  Title VARCHAR(200) NOT NULL,  
  ReleaseDate DATE,  
  GenreID INT NOT NULL,  
  CONSTRAINT FK_Movie_Genre  
    FOREIGN KEY (GenreID)  
    REFERENCES Dim_Genre(GenreID)  
);
```

8 COMPOSE SQL STATEMENTS

K&Ps: P9, P10 (Part 1); K2, P2, P4 (Part 2)

SELECT STATEMENT (DML)

```
SELECT m.Title, g.GenreName, m.ReleaseDate  
FROM Movie m  
JOIN Dim_Genre g ON m.GenreID = g.GenreID  
WHERE m.ReleaseDate >= '2020-01-01'  
ORDER BY m.Title;
```

RESULTS

Title	GenreName	ReleaseDate
Dune: Part Two	Sci-Fi	2024-03-01
Oppenheimer	Drama	2023-07-21
The Batman	Action	2022-03-04
...	...	...

9 CREATE A BASIC SECURITY PLAN

K&Ps: K12, K13, P2, P11 (Part 1);
- K14, K15, P5 (Part 2)

- Screenshots of Systems Administrator areas
- Identify user roles
- Define access levels
- User authentication
- Authorisation & permissions
- Data encryption
- Audit trails & logging
- Regular user access reviews (optional)
- Training & awareness (optional)

USER ROLES

ROLE	ACCESS LEVEL
DBA	Full Access
Data Analyst	Read/Query
Data Editor	Create/Update
Read Only User	Read Only



10 CREATE A BASIC GOVERNANCE PLAN

K&Ps: P9, P11 (Part 1); K3, K16, P6, P7 (Part 2)

- Data quality assurance
- Data update mechanism
- Right to edit
- Data removal mechanism
- Privacy & consent management
- Data retention policies (optional)
- Documentation & communication (optional)
- Regular review & compliance monitoring (optional)

11 ASSESS THE PROS AND CONS OF BIG DATA SYSTEMS & DATA WAREHOUSING

K&Ps: K12, K13 (Part 2)



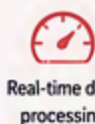
Dataset characteristics



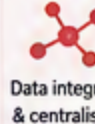
Scalability requirements



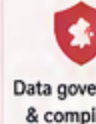
Complex data processing



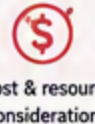
Real-time data processing



Data integration & centralisation



Data governance & compliance



Cost & resource considerations



Analytics & insights

BIG DATA SYSTEMS (e.g., Hadoop, Spark)

PROS

- Handles large volume, variety & velocity
- Highly scalable
- Cost-effective on commodity hardware
- Good for unstructured/semi-structured data
- Real-time & advanced analytics capable

CONS

- Complex to implement & maintain
- Requires specialised skills
- Data quality can be inconsistent
- Higher operational overhead
- Not ideal for complex SQL queries

DATA WAREHOUSING (e.g., SQL DW, Snowflake)

PROS

- Optimised for SQL & reporting
- High data quality & consistency
- Strong governance & security
- Faster query performance for analytics
- Easier to integrate structured data

CONS

- Higher cost
- Less flexible with unstructured data
- Batch-oriented (not real-time)
- Scaling can be more expensive
- Requires upfront data modelling



A well-designed data solution follows a lifecycle: Plan → Design → Build → Manage → Evaluate.



Good luck!